

Output Ontology (OO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: HOPE | Context: US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The 12-label taxonomy is explicitly 'aligned with mental-health counseling session' [Q17, Q104] and was designed in consultation with therapists [Q27]. The user judges the taxonomy fit-for-purpose for conversational flow tracking (elicitation A3). However, internal taxonomy stress is documented: ACK underperforms at F1=47.86% [Q77], indicating semantic overload even within clinical scope; confusion analysis shows YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%) error rates [Q91]; complementary label pairs are 'annotation-boundary-challenging' [Q51, Q52]. For the deployment's peer support and crisis extensions, the GC catch-all [Q44] becomes the de facto repository for any communicative act outside the clinical frame, and crisis-specific acts (safety-planning, risk stratification) lack dedicated labels — collapsing into IRQ/YNQ with potentially safety-critical consequences. The discarded emotion annotation pilot [Q126, Q127, Q129] indicates the authors recognized affective dimensions but chose not to incorporate them. Output is single-label per utterance [Q30], which is appropriate for the framing but limits expressiveness for utterances spanning multiple acts. Score of 3 reflects fit-for-purpose taxonomy for in-scope clinical use but real boundary ambiguity at deployment edges.

ID	Evidence
Q17	"We propose twelve dialogue-act labels that are aligned with mental-health counseling session." (p.2)
Q27	"Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3)
Q30	"Sometimes, an utterance can have multiple dialog-acts; however, they are rare in the annotated dataset. Hence, for simplicity, we consider only one (primary) label for each utterance." (p.3)
Q44	"General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation." (p.4)
Q51	"The broad definitions of speaker-initiative and speaker-responsive dialogue-act label pairs, IRQ & ID; ORQ & OD, and CRQ & CD, seem complementary in nature." (p.4)
Q52	"However, the dataset does not support the above view entirely." (p.4)
Q77	"We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%." (p.7)
Q91	"We observe three pairs with significant error-rates ($\geq 25\%$) – YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%)." (p.7)
Q104	"We defined twelve dialogue-act labels to cater to the requirement of counselling sessions." (p.8)
Q126	"In earlier phases of experiments, we also considered the role of emotion in dialogue act classification." (p.11)
Q127	"We annotated a subset of our dataset with three emotion classes consisting of 'positive', 'negative' and 'neutral'." (p.11)
Q129	"Due to such imbalance, this data was not used in the final version of our proposed architecture." (p.11)

Strengths

- Taxonomy designed in consultation with therapists and counseling experts [Q27], grounded in counseling-session requirements rather than borrowed
- Three-tier hierarchical structure (initiative/responsive/general) [Q29, Q32] provides interpretable organization

- Per-label F1 reporting [Q76] surfaces label-level performance limitations transparently
- Authors explicitly disclaim diagnostic intent [Q102], appropriate for a flow-classification scope

ID	Evidence
Q27	"Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3)
Q29	"Each utterance in the dialogue belongs to one of the three categories – speaker initiative, speaker responsive, and general." (p.3)
Q32	"Our annotation scheme assigns three distinct dialogue-act labels to the first two categories, while the remaining four labels belong to the general category." (p.4)
Q76	"We also present the label-wise performance of SPARTA in Table 4." (p.7)
Q102	"It is important to note that we do not make any diagnostic claims." (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Labels are nominally regionally relevant (greeting, information delivery, etc. are universal) but encode US clinical session structure assumptions [Q98]. South Asian directiveness norms may shift label-boundary behavior.

OO-2: Missing categories: (a) crisis-specific safety-planning and risk-assessment acts; (b) peer support emotional co-regulation acts; (c) empathic reflection vs. simple acknowledgment distinction (the user notes this is an intentional out-of-scope choice — A3); (d) affective dimension was piloted but discarded due to imbalance [Q126, Q127, Q129].

OO-3: GC as a residual catch-all [Q44] effectively encodes 'not part of the structured clinical frame' as a category, which biases against acts characteristic of peer/crisis registers. Speaker asymmetry [Q125] presupposes therapist-patient hierarchy not present in peer dyads.

OO-4: Stakeholder-driven taxonomy redesign is needed for crisis-context deployment specifically; for in-scope clinical and South Asian extensions, the user judges core labels acceptable (A3).

OO-5: Documented issues: ACK overload [Q77], complementary-pair confusion [Q51, Q52, Q91], lack of crisis or peer support labels, abandoned emotion dimension, and the GC residual category becoming a semantic black hole for out-of-frame utterances.

ID	Evidence
Q17	"We propose twelve dialogue-act labels that are aligned with mental-health counseling session." (p.2)
Q27	"Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3)
Q44	"General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation." (p.4)
Q51	"The broad definitions of speaker-initiative and speaker-responsive dialogue-act label pairs, IRQ & ID; ORQ & OD, and CRQ & CD, seem complementary in nature." (p.4)

Q52	"However, the dataset does not support the above view entirely." (p.4)
Q77	"We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%." (p.7)
Q91	"We observe three pairs with significant error-rates ($\geq 25\%$) – YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%)." (p.7)
Q98	"Another important aspect of the current work is that the majority of the sessions in HOPE belong to the mental health professionals and patients based in United States." (p.8)
Q102	"It is important to note that we do not make any diagnostic claims." (p.8)
Q125	"This is particularly apparent as a majority portion of the counseling conversation is driven by the therapist, thus the speaker-initiative dialogue-acts have higher relevance with the therapist utterances." (p.11)
Q126	"In earlier phases of experiments, we also considered the role of emotion in dialogue act classification." (p.11)
Q127	"We annotated a subset of our dataset with three emotion classes consisting of 'positive', 'negative' and 'neutral'." (p.11)
Q129	"Due to such imbalance, this data was not used in the final version of our proposed architecture." (p.11)
WEB-2	988 Lifeline three-phase crisis model uses communicative acts not represented in HOPE labels (https://988lifeline.org/wp-content/uploads/2023/02/FINAL_988_Suicide_and_Crisis_Lifeline_Suicide_Safety_Policy_-3.pdf)
WEB-18	ESConv 8-strategy framework for peer support is structurally distinct from HOPE (https://aclanthology.org/2021.acl-long.269/)

Information Gaps

- Per-label inter-annotator agreement breakdown (only aggregate Kappa reported)
- How GC and ACK absorb peer/crisis acts in practice — no empirical validation

Requires Expert Verification

- Whether crisis intervention staff can operate with IRQ/YNQ collapse for safety-planning probes
- Whether peer support practitioners would adopt the speaker-asymmetric taxonomy

Remediation

Gap 1: ACK label is semantically overloaded (F1=47.86%) [Q77] and GC functions as a residual catch-all [Q44]; per-label and per-session-type Kappa breakdowns are unavailable.

Recommendation: Request or compute per-label and per-session-type inter-annotator agreement on a re-annotated subset. For deployment, suppress or flag (rather than display) ACK and GC predictions in supervisory dashboards until reliability is independently validated.

ID	Evidence
Q44	"General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation." (p.4)
Q77	"We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%." (p.7)